NuKAT Mission Definition Review

October 15th, 2025

Mission Advisory Board

- Dave Copeland
  - Space Communications System Engineer, John Hopkins University Applied Physics Laboratory
  - Co-Founder, Silversat
- Chris Hersman
  - Mission Systems Engineer, John Hopkins University Applied Physics Laboratory
- Glen Fountain
  - Project Manager, John Hopkins University Applied Physics Laboratory – Retired
  - Former Industry Advisor, Mike Wiegers Department of Electrical and Computer Engineering, Kansas State University

Mission Advisory Board

- Dr. Tim Steffensmeier
  - Assistant Vice President and Director of Engagements and Outreach, Kansas State University
- Bill Swartz
  - Atmospheric and Remote Sensing Scientist, John Hopkins University Applied Physics Laboratory
- Dr. Amir Bahadori
  - Professor and Program Director, Kansas State University Nuclear Engineering
  - Former NASA civil servant in the Space Radiation Analysis Group (SRAG)

CubeSat Project Donor

- Kevin Needham
  - President, Kiewit Power Engineering
  - Senior Vice President, Power Markets & Strategy

Agenda

Team Structure
- Majors and Locations
- Sub-Team Grouping

Mission & System Overview
- Mission Overview *Colby Johnston*
- System Overview *Bailey Walke*

Technical Implementation
- Subsystem Requirements *Sub-Team Leads*
- Ground Station Overview *Noah Amundson* *Andrew Osterhaus*

Managerial
- Integration & Testing *Chase Newell*
- Budget & Schedule *Abigail Husted*
- Risk Assessment *Owen Franken*

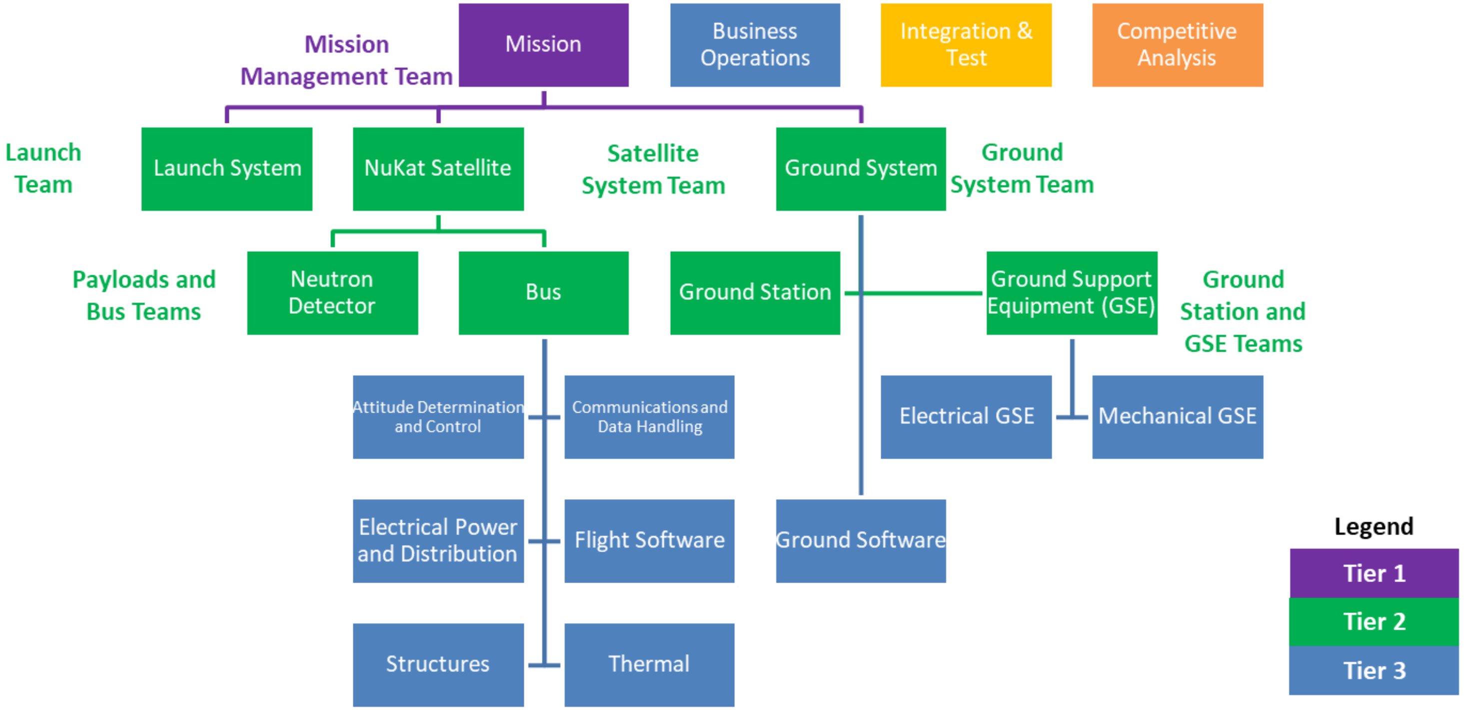
Conclusion
- Path to PDR *Bailey Walke*
- Q/A

Team Acknowledgement

- 23 students (6 majors)
  - 2 campuses
    - Manhattan
      - Mechanical Engineering
      - Nuclear Engineering
      - Electrical Engineering
      - Computer Science
      - Industrial Engineering
    - Salina
      - Mechanical Engineering Technology

- 2 Faculty Advisors
  - Prof. Paul Snider
  - Dr. Raju Dandu

Team Structure



Purpose of MDR

Objective

Define mission concepts, objectives, and system-level requirements

Verify subsystems and operation concepts align with mission goals

Outcomes

- Confirmation of technical feasibility
- Definition of mission requirements and architecture
- Identification of major risks and mitigation strategies

Review Focus Areas

- Mission objectives and rationale
- System concept and interfaces
- Orbit and environment assumptions
- Schedule, budget and risk strategies
- Readiness for path forward to design review

MISSION OVERVIEW

Colby Johnston – NuKAT Principal Investigator

Mission Background

Common Neutron Space Detectors	Downsides
$^3He/BF_3$ Proportional Gas	Pressure Vessel
Scintillators	Large Crystals/Photomultiplier tubes
Semiconductors (e.g., Si(Li))	Limited efficiency from planar geometry

- Microstructured Semiconductor Neutron Detectors (MSNDs)
- Original Domino heritage
  - NeRDI-1A and -1B missions
- New DSC Domino

Neutron Detection Metrics
- Flux
- Energy Spectrum
- Dose or Dose Rate

Payload Details

- Four detector packages
  - Two primary and two backup
- Four coarse energy bins
  - MSND Backfill (^{6}LiF and hydrogenous)
  - Moderation (HDPE)

Unmoderated
Thermal & Fast Energy

- Primary (DSC-0)
- Backup (DSC-1)

Moderated
Epithermal & Fast Energy

- Primary (DSC-2)
- Backup (DSC-3)

Mission Objectives

- DSC Domino space qualification
- Provide neutron counts, energy bins, and orbital location
- K-State CubeSat program pathfinder

Mission Statement

Advancing space-based neutron detection via microstructured semiconductor neutron detectors (MSNDs) in low Earth orbit and validating the DSC Domino as a space-qualified detector platform.

Mission Product

Stakeholders

Tertiary
- National Laboratories*
- NOAA*

Secondary
- Other Universities*
- NASA's Space Radiation Analysis Group (SRAG)

Core
- K-State
- Radiation Detection Technologies (RDT)

End Products
Background radiation rejection scheme
Coarse energy histogram of neutron environment
Heat map of neutron flux in energy bins
DSC Domino flight heritage and space qualification

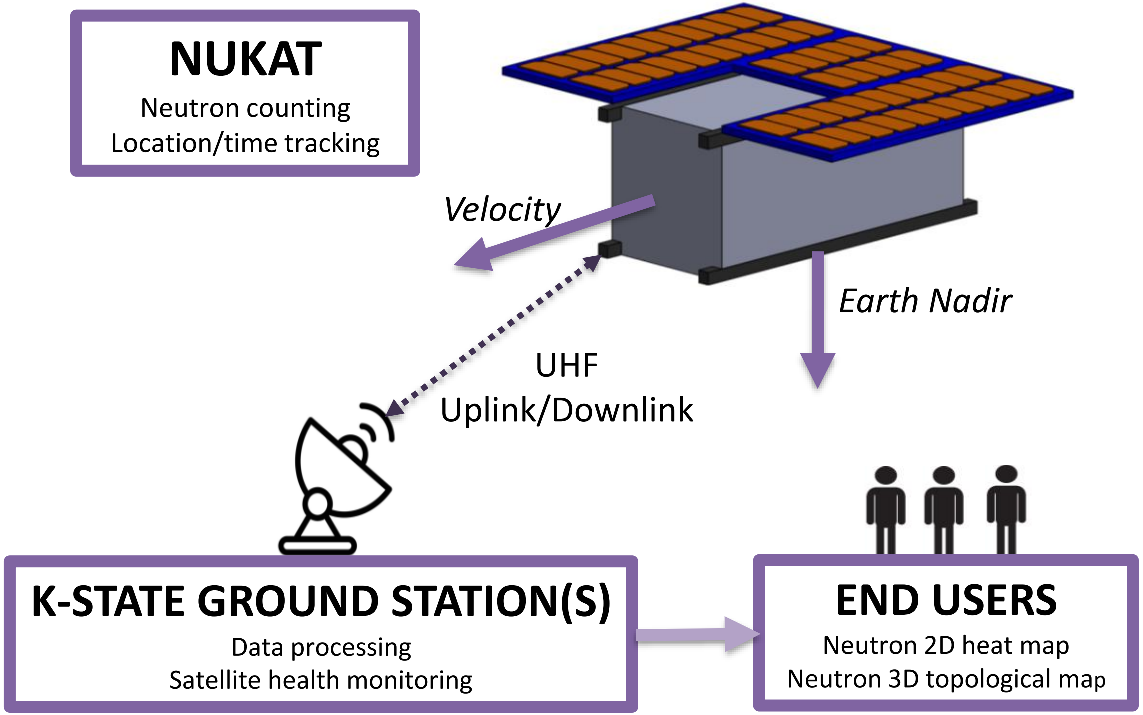
* Ongoing communication establishing these stakeholders

SYSTEM OVERVIEW

Bailey Walke – NuKAT Chief Systems Engineer

System Requirements/Mission Block Diagram

Requirements
Neutron detector measures neutron counts and energy bins
Locational and time trace data
Survives launch and on-orbit environments
Uplink/Downlink capability with ground station
Operational Optimization



Concept of Operations

LAUNCH & INITIALIZATION

1 Month

1. Deployment from LV
2. Sleep
3. Initial Startup
4. Detumble
5. Appendages Deploy
6. Calibration
7. Stand By

MISSION OPERATIONS

12 Months

Science

- Counting
- Pointing
- Receiving
- Transmitting

Safe

- Detumble
- Over/under Voltage
- EMI Survival
- Over/under Temperature

Emergency

DECOMISSIONING

6-12 Months

1. Increased Drag Positioning
2. Final Transmission
3. Loss of Communication
4. Burn Up

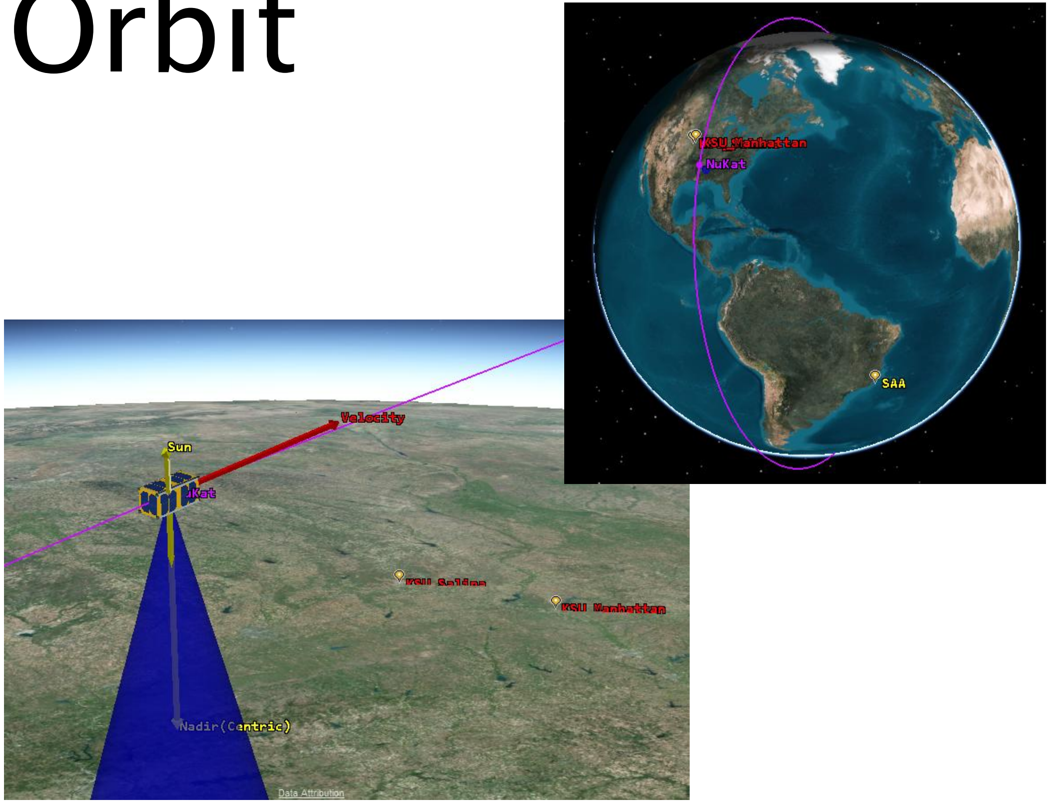
ORBIT & ENVIRONMENT

Phillip Shirkey, Igor Sheshukov

Orbit

Points of Interest	
Ground Stations	**Science**
KSU Manhattan	South Atlantic Anomaly (SAA)
KSU Salina	Earth Poles

Orbit Range	
Altitude	LEO (350km-600km)
Inclination	Polar (~75°-114°)
Type	Circular



Orbit

- Lifetime
- Downlink
- Revisit
- Antenna
- Ground Stations

Assumptions: 5kg mass, 15° antenna half angle, fixed antenna, 395km alt., 108° inc.	
Lifetime	3.0 years
Downlink Time	2-30sec; avg 24sec
Revisit Time	2.5 or 5.5 days

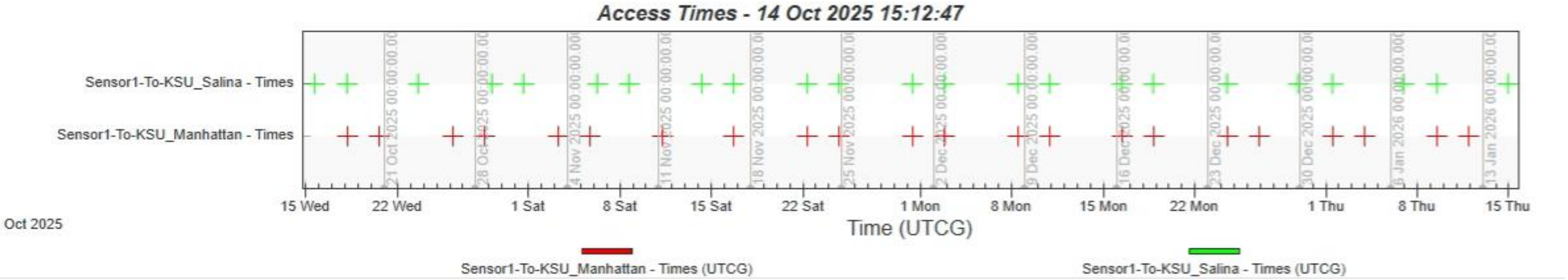
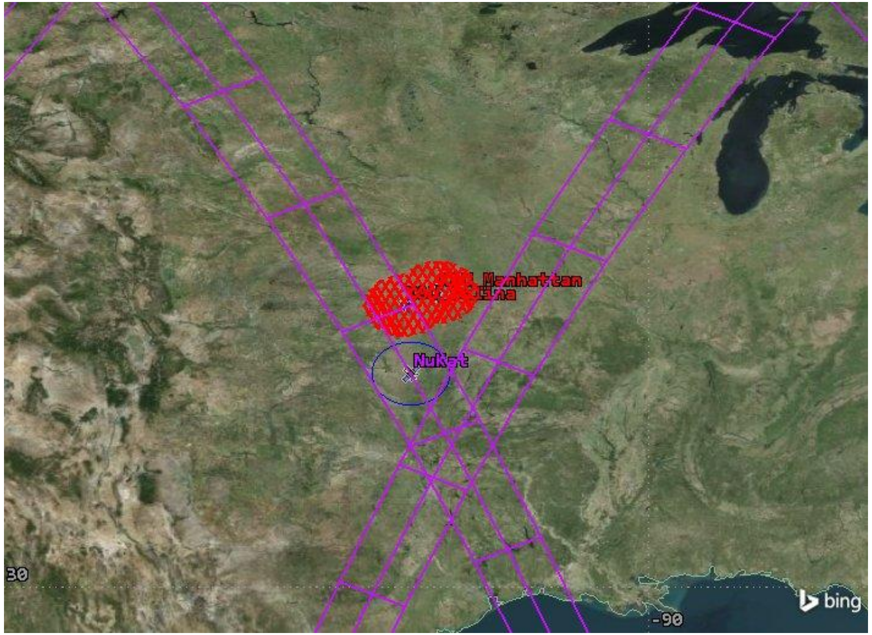
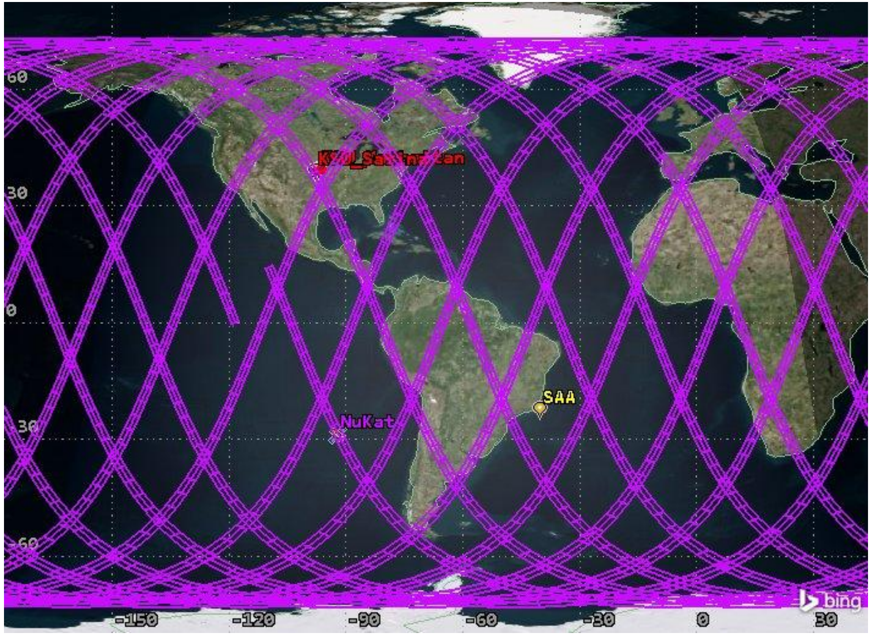


Figure: Revisit times for a polar orbit

Orbit





Left Figure: Zoomed in view of polar orbit over KSU
Right Figure: Zoomed out view of polar orbit over two days

Space Environment

- Thermal Cycling
- Radiation
- Debris
- Solar and Electromagnetic Events
- Atmosphere

Launch Environment

- Shock & Vibrations
- Acceleration
- Electromagnetic Compatibility
- Ascent Venting
- Other Deployment Constraints

SATELLITE SUBSYSTEMS

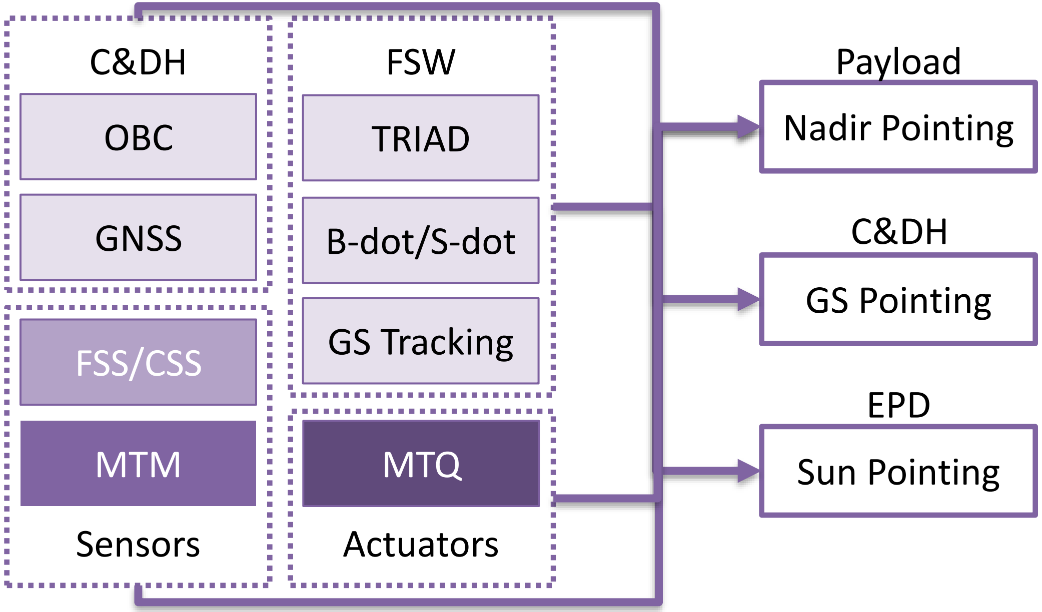
Phillip Shirkey, Logan Hayward, Michael Bennett, Braden Adams, Sophia Fenton, Drew Sallman

ADCS

Driving Requirements

1. Nadir Pointing
2. Ground Station Pointing
3. Sun Pointing

Block Diagram

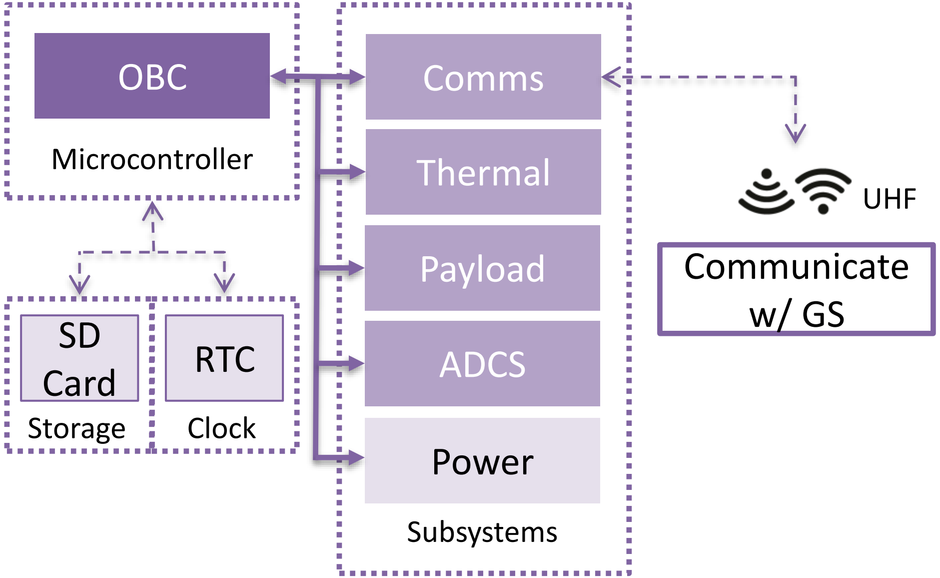


C&DH

Driving Requirements

1. UHF link (Tx/Rx)
2. Interface w/ Subsystems (MSN)
3. Radiation Tolerance
   - ❖ Error Correcting
   - ❖ Fault Detection
   - ❖ Robustness

Block Diagram

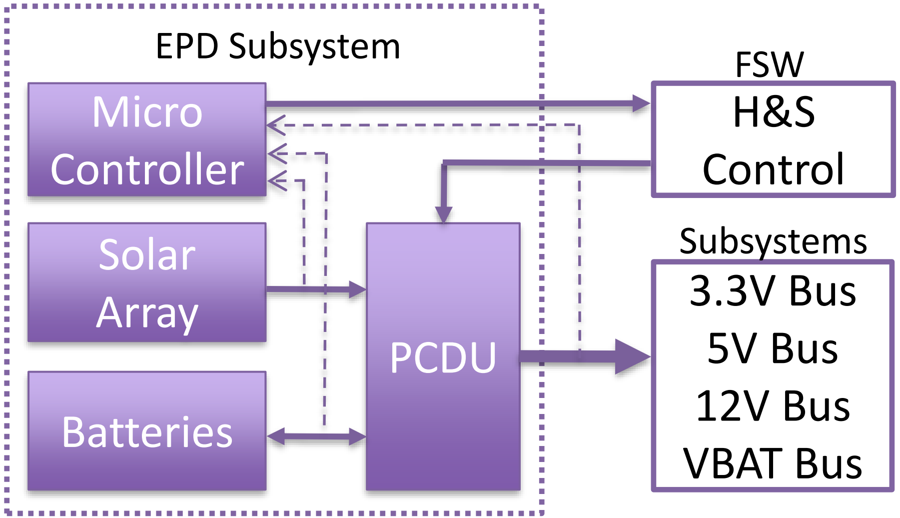


EPDS

Driving Requirements

1. Remaining Power Positive
2. Ability to Test & Simulate
3. Ability to Monitor Current & Voltage

Block Diagram

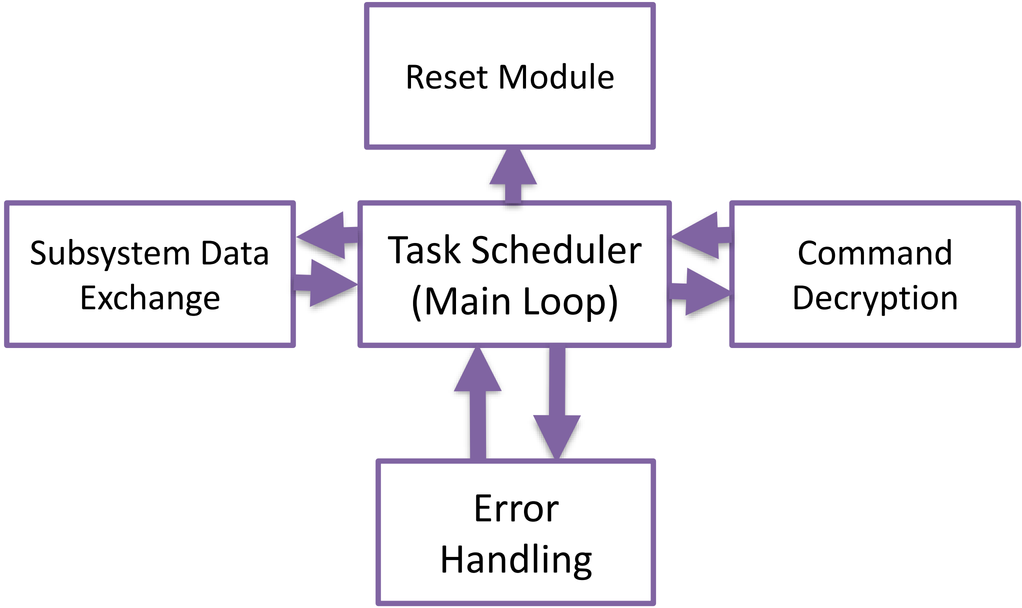


FLIGHT SOFTWARE

Driving Requirements

1. Respond to various operational modes
2. Collect and report subsystem telemetry data
3. Respond to detected errors

Block Diagram

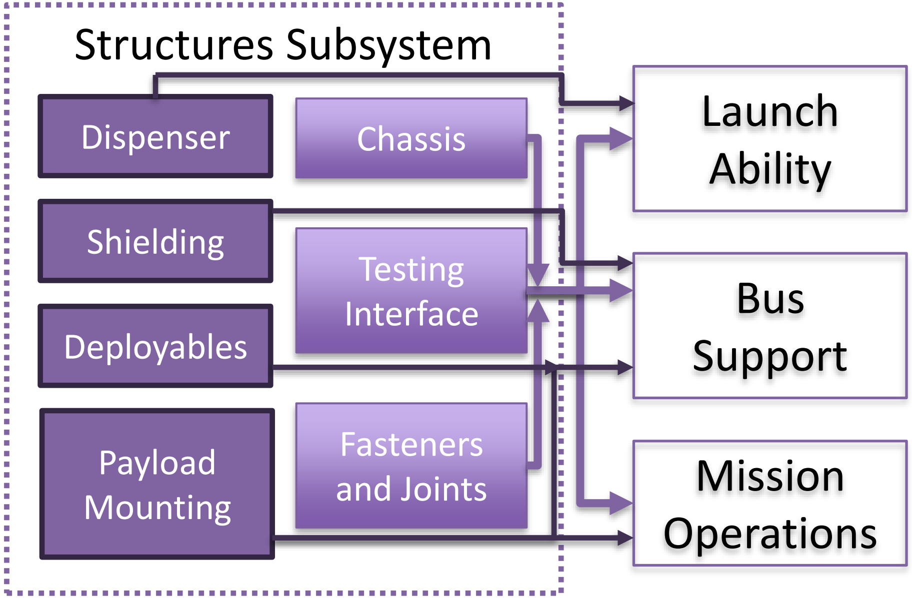


STRUCTURES

Driving Requirements

1. Launch Compatibility and Survivability
2. Satellite Bus Support
3. Mission Operations

Block Diagram

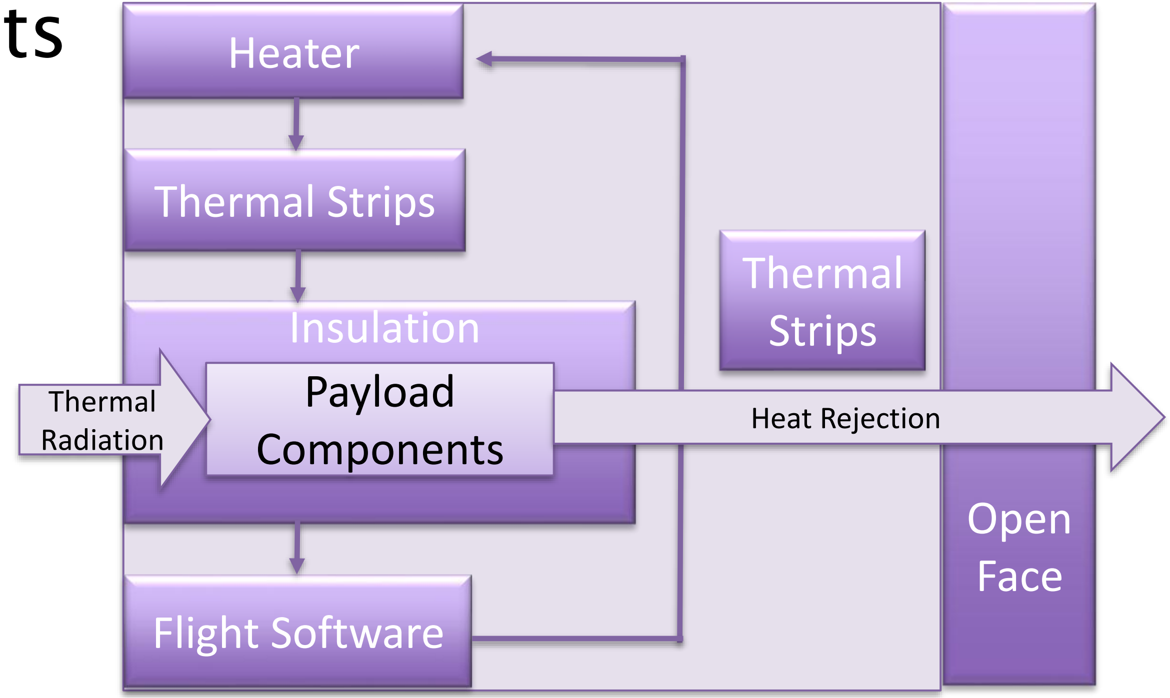


Sophia Fenton

THERMAL Block Diagram

Driving Requirements

1. Maintain component operable temperature range
2. Maintain a stable/non-fluctuating temperature range (insulation, flight software, PID)



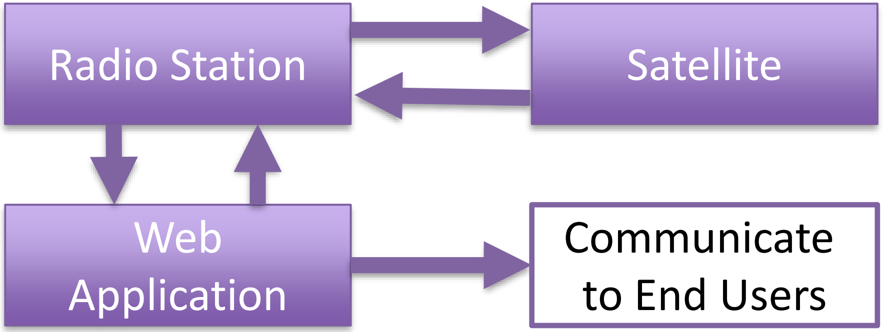
GROUND STATION

Noah Amundson, Andrew Osterhaus

Ground Station

Driving Requirements

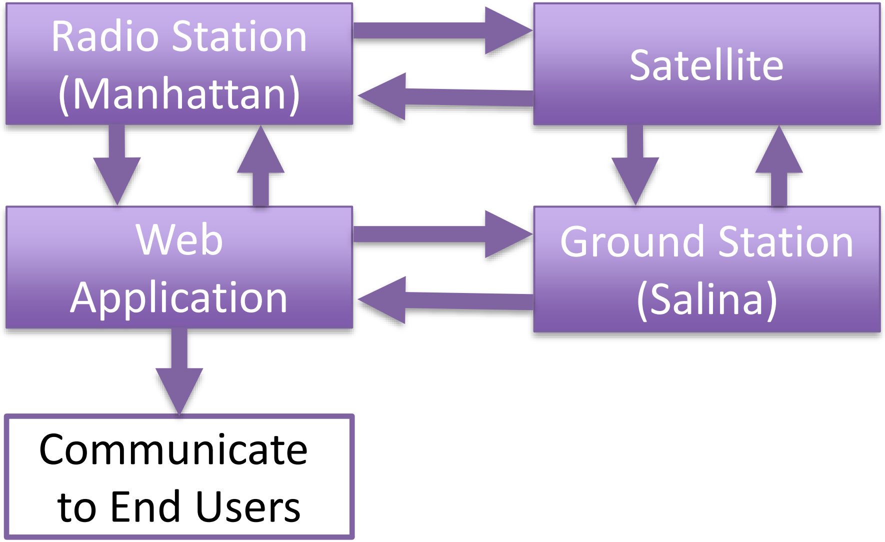
1. Communicate with satellite between 420-450 MHz
2. Ground system must be able to track NuKAT in orbit
3. Uplink and downlink data between ground station and NuKAT



Ground Station Hardware

Driving Requirements

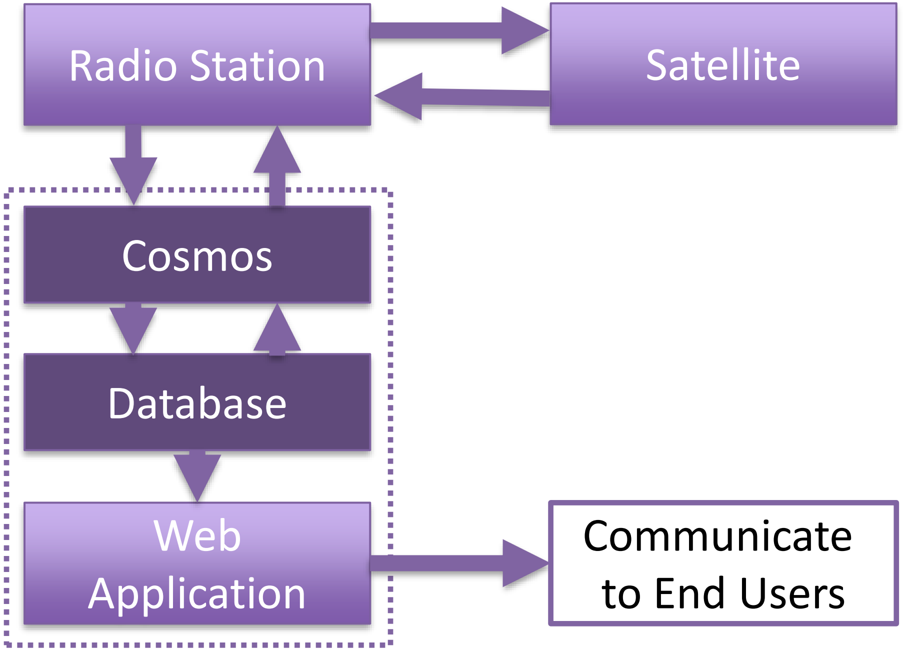
1. The ground station shall have a primary and secondary location.
2. Ground system shall meet the power and lifetime requirements of the mission.
3. Personnel must have training to operate ground station.



Ground Station Software

Driving Requirements

- Communicate with the satellite
    - Receiving downlink data
    - Up-link commands
- Maintain status of the satellite
- Create meaningful information for the end users



Integration and Test

- RVM Verification Methods
  - Grouped by Analysis, Test, Inspect, Demonstrate (ATID)

Mission Parameter (Requirements)	Type	Description
Neutron Detector Payload Accuracy	D	Demonstrate that DSC Domino neutron detector is accurate within "x" percent
Core Capabilities (Nadir Tracking)	T	Test that Attitude Determination system can track the Nadir datum
Operational Lifetime (Attitude Determination)	A	Analyze ADCS in environmental conditions of space for "x" years.
Neutron Detector Mounting Angle	I	Inspect the detector mounts and ensure correct orientation.

  - 169 of 213 evaluated, expected completion Oct 16th, 2025
- MGSE/EGSE Equipment (May Change)
  - Multimeter, power supply, function generator, oscilloscope, data acquisition system, RF-SCOE device, and grounding mat

BUDGET & SCHEDULE

Abigail Husted

Budget - Hardware

- A more detailed Excel Spreadsheet for budget tracking when individual parts are ordered
- Broad categories based on NASA CubeSat 101, Cal Poly Phat Sat (3U) and Global Spec CubeSat articles

Item	Cost
Payload	~$20,000
EPDS	~$10,000
ADCS (from NuKat)	$9,600
Contingency (leftover)	$9,300
C&DH	~$2,700
On Board Computer	~$5,500
Structure	~$4,000

Budget – Hardware Cont.

Item	Cost
Harnessing & Connectors	~$1,500
Thermal	~$1,000
Flight Software	~$6,000
Integration	~$6,000
Testing	~$6,000
Ground Station	~$3,000
Spares	~$2,400

~$100,000 total

Abigail Husted

Schedule

- Gantt charts and MS Planner to manage
- Mostly TBD, with more details to come after RVM is complete
  - Development – 12-24 months avg
  - Launch contracting – 6-18 months avg
  - Integration – 3-9 months avg
  - Launch Queue – 3-12 months avg
  - **<u>University CubeSat total– 18-24 months</u>**

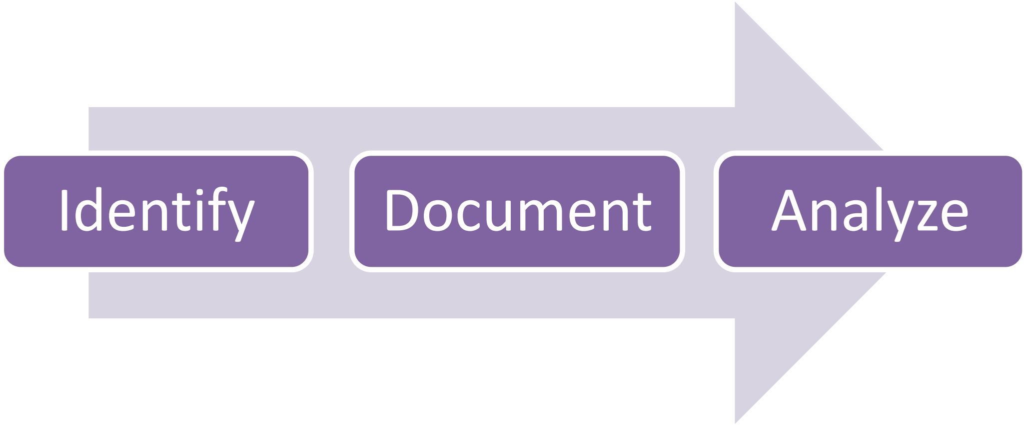
Abigail Husted

RISK ASSESSMENT

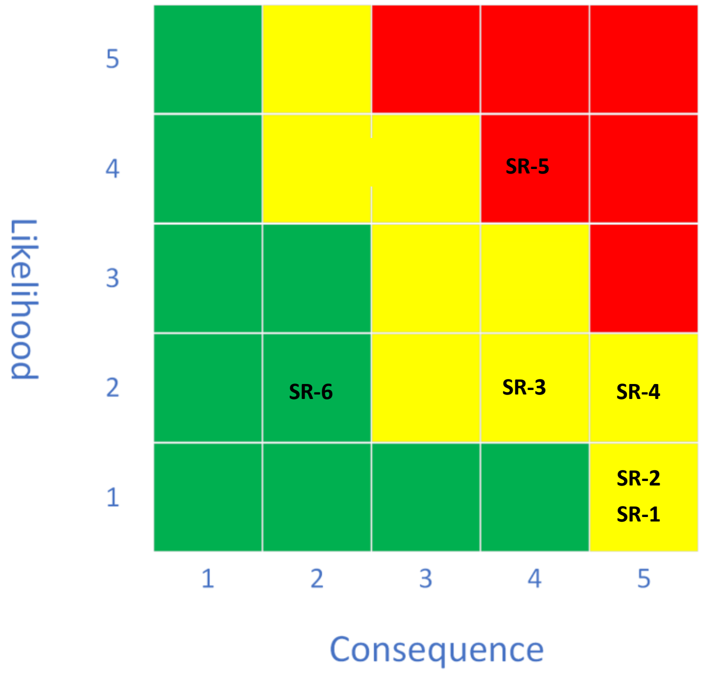
Risk Assessment Strategy

Driving Requirements

- Creating stable and accurate simulations to mitigate risk
- Consider all potential areas of risk
- Ensure risk considers specifications



System-Level Risks



ID	Risk	Mitigation Strategy	Approach
SR-1	Not surviving launch environment	Extensive testing to selected launch vehicle specifications	M
SR-2	Not surviving low earth orbit space environment	Use space rated hardware and test hardware at both component and system configuration	M
SR-3	Not deploying from launch vehicle	Strict compliance with design and material specification of selected launch provider	M
SR-4	Deployable system failure	Design system to survive non-deployment of solar panels/antennas. Redundant design of deployables	M
SR-5	High radiation environment	Simulate radiation environments to determine potential impacts	M
SR-6	Lack of knowledge transfer/project turnover	Prioritizing documentation, recruitment, and development of younger students to become team leads	M

CONCLUSION

Bailey Walke

Path to PDR

Focus Area	PDR Goal
Orbit & ConOps	Finalize orbit selection, complete operations timeline, and verify feasibility with ground passes.
Requirements	Finalize system and subsystem requirements; ensure all are measurable and verifiable in the RVM.
Architecture & Interfaces	Freeze top-level architecture; define mechanical, electrical, and data interfaces.
Preliminary Subsystem Designs	Preliminary CAD, thermal models, power budgets, and link budgets.
Risk Reduction	Mitigate system-level risks, develop sub-system risks and assess
Verification & Test Planning	Outline environmental test plan per GEVS and identify required facilities.

NuKAT Design Review to be presented later this semester

MAB Feedback

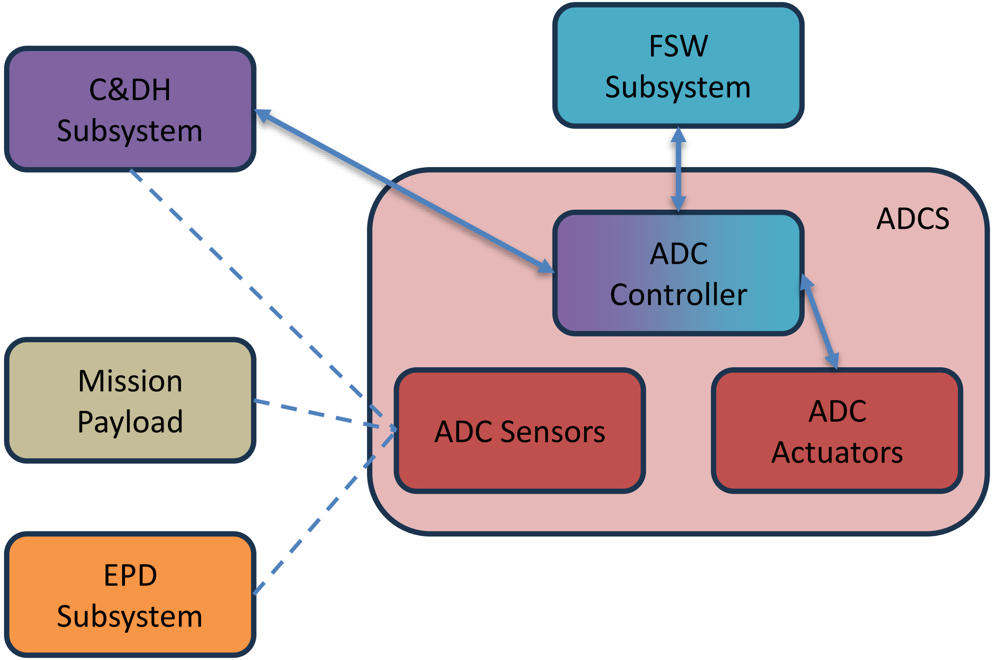
- What mission definition content is missing?
- What mission definition needs more development?
- What mission definition trade studies are needed?

BACKUP

Presenter List

Section	Presenter	Time	Section	Presenter	Time
Introduction	Bailey	3.25	STR	Sophia	1.5
Mission Background	Colby	4.5	THM	Drew	1.5
Mission Overview	Bailey	4.5	GS (added slide ^)	Noah	1
Orbit/Environments	Phillip & Igor	3-3	GSW	Andrew & Nalen	1
ADCS	Phillip	1.25	I&T	Chase	1
CDH	Logan	1.5	Risk	Owen	1.25
EPD	Mike	1.5	Schedule/Budget	Abigail	1
FSW	Braden & Nalen	1.5	Conclusion	Bailey	1.25

Attitude Determination and Control System



ADCS Subsystem Core Components

Sensors
- Fine & Coarse Sun Sensors
- Magnetometer
- Navigation Satellite Receiver

Actuators
- Magnetorquers

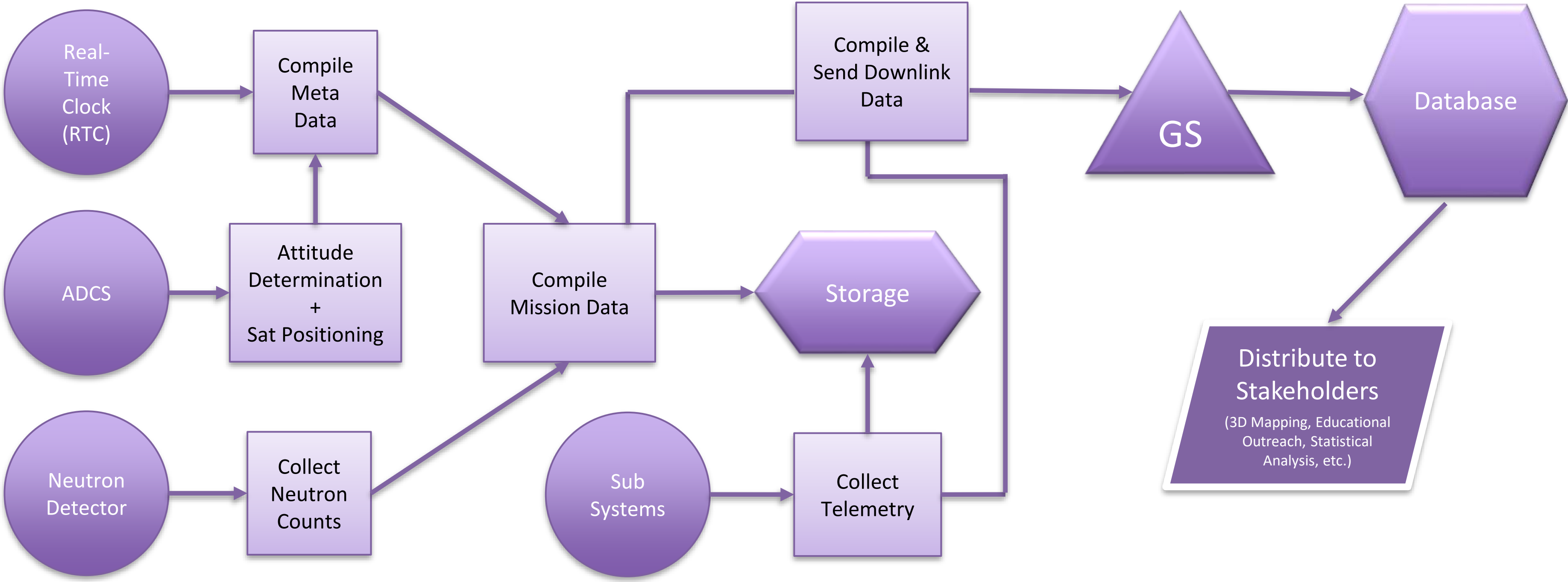
Computer
- Onboard Computer

Software
- Angle and Spin Determination
- Angle and Spin Control
- Ground Station Tracking

ADCS Critical Tech — COTS or Custom

ADCS Critical Tech	COTS or Custom
Fine Sun Sensor 1-6x	Custom – Cost, Precision
Coarse Sun Sensor 6x	Custom – Cost, Precision
Triaxial Magnetometer	COTS – Precision
Navigation Satellite Receiver	COTS – Precision
Magnetorquer 3x	COTS – Precision
Onboard Computer	COTS/Custom – Cost, Precision
TRIAD Angle Algorithm	Custom – Cost
B-Dot Spin Algorithm	Custom – Cost
GS Tracking Algorithm	Custom – Cost
General Flight Algorithm	Custom – Cost
Upset Recovery Algorithms	Custom – Cost

Data Flow Diagram



NOTE: This is VERY high level. Each square can be broken into its own diagram.

Acronyms

- ADCS – Attitude Determination and Control System
- ATID – Analysis, Test, Inspect, Demonstrate
- B-dot/S-dot – Method of angular velocity determination using change in Earth or Sun vector orientation over time
- CAD – Computer Aided Design
- C&DH – Communications and Data Handling
- ConOps – Concept of Operations
- CSS – Coarse Sun Sensor
- EGSE – Electrical Ground Station Equipment
- EM – Electromagnetic
- EMI – Electromagnetic Interference
- EPD – Energy and Power Distribution
- FSS – Fine Sun Sensor
- FSW – Flight Software
- GEVS – General Environmental Verification Standard
- GNSS - Global Navigation Satellite System
- GS – Ground Station
- GSE – Ground Support Equipment
- HDPE – High Density Polyethylene
- H&S – Health and Status
- KSU – Kansas State University
- LEO – Low Earth Orbit
- LV – Launch Vehicle
- MAB – Mission Advisory Board
- MDR – Mission Definition Review
- MGSE – Mechanical Ground Station Equipment
- MS – Microsoft
- MSN – Main Subsystem Network
- MSND – Microstructured Semiconductor Neutron Detector
- MTM – Magnetometer
- MTQ – Magnetorquer (either air or solid core)
- NASA – National Aeronautics and Space Administration
- NOAA – National Oceanic and Atmospheric Administration
- OBC – Onboard Computer
- PCDU – Power Control Distribution Unit
- PDR – Preliminary Design Review
- PID – Proportional-Integral-Derivative (controller)
- RDT – Radiation Detection Technologies
- RF-SCOE – Radio Frequency Special Check-Out Equipment
- RTC – Real Time Clock
- RTD – Resistance Temperature Detector
- RVM – Requirement Verification Matrix
- Rx – Receive
- SAA – South Atlantic Anomaly
- SR – System Risk
- TBD – To Be Determined
- TRIAD – Tri-Axial Attitude Determination
- Tx – Transmit
- UHF – Ultra High Frequency